admission that the program can't stand on its own two feet, cannot be justified by the purpose for which it was originally sold.

Once upon a time it was thought, on the basis of econometric models, that for every dollar invested in NASA many dollars were pumped into the U.S. economy. If this multiplier effect applied more to NASA than to most government agencies, it would provide a potent fiscal and social justification for the space program. NASA supporters were not shy about appealing to this argument. But a 1994 Congressional Budget Office study found it to be a delusion. While NASA spending benefits some production segments of the U.S. economy—especially the aerospace industry—there is no preferential multiplier effect. Likewise, while NASA spending certainly creates or maintains jobs and profits, it does so no more efficiently than many other government agencies.

Then there's education, an argument that has proved from time to time very attractive in the White House. Doctorates in science peaked somewhere around the time *of Apollo 11*, maybe even with the proper phase lag after the start of the *Apollo* program. The cause-and-effect relationship is perhaps undemonstrated, although not implausible. But so what? If we're interested in improving education, is going to Mars the best route? Think of what we could do with $100 billion for teacher training and salaries, school laboratories and libraries, scholarships for disadvantaged students, research facilities, and graduate fellowships. Is it really true that the best way to promote science education is to go to Mars?

Another argument is that human missions to Mars will occupy the military-industrial complex, diffusing the temptation to use its considerable political muscle to exaggerate external threats and pump up defense funding. The other side of this particular coin is that by going to Mars we maintain a standby technological capacity that might be important for future military contingencies. Of course, we might simply ask those guys to do something directly useful for the civilian economy. But as we saw in the 1970s with Grumman buses and Boeing/Vertol commuter trains, the aerospace industry experiences real difficulty in producing competitively for the civilian economy. Certainly a tank may travel 1,000 miles a year